Data Cleaning and Preparation

Problem Statement: Analyzing Customer Churn in a Telecommunications Company Dataset: "Telecom_Customer_Churn.csv" Description: The dataset contains information about customers of a telecommunications company and whether they have churned (i.e., discontinued their services). The dataset includes various attributes of the customers, such as their demographics, usage patterns, and account information. The goal is to perform data cleaning and preparation to gain insights into the factors that contribute to customer churn. Tasks to Perform:

1. Import the "Telecom_Customer_Churn.csv" dataset.
2. Explore the dataset to understand its structure and content.
3. Handle missing values in the dataset, deciding on an appropriate strategy.
4. Remove any duplicate records from the dataset.
5. Check for inconsistent data, such as inconsistent formatting or spelling variations, and standardize it.
6. Convert columns to the correct data types as needed.
7. Identify and handle outliers in the data.
8. Perform feature engineering, creating new features that may be relevant to predicting customer churn.
9. Normalize or scale the data if necessary. Faculty of Science and Technology Savitribai Phule Pune University Syllabus for Fourth Year of Artificial Intelligence and Data Science (2020 Course) 47/126
10. Split the dataset into training and testing sets for further analysis.
11. Export the cleaned dataset for future analysis or modeling.

Code :

1. Import the "Telecom_Customer_Churn.csv" dataset.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Telco_Customer_Churn_Dataset.csv")
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No

5 rows × 21 columns

2. Explore the dataset to understand its structure and content.

df.shape

(7043, 21)

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64 
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64 
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
```

df.describe()

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

df.columns

```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

df.drop('customerID',axis='columns',inplace=True)

df.dtypes

```
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges         object
Churn                object
dtype: object
```

3. Handle missing values in the dataset, deciding on an appropriate strategy.

df.isnull().sum()

```
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges         0
Churn                0
dtype: int64
```

4 . Remove any duplicate records from the dataset.

df.duplicated()

```
0       False
1       False
2       False
3       False
4       False
        ...
7038    False
7039    False
7040    False
7041    False
7042    False
Length: 7043, dtype: bool
```

5. Check for inconsistent data, such as inconsistent formatting or spelling variations,and standardize it.

df.TotalCharges.values # it is in string format ---> convert to float

```
array(['29.85', '1889.5', '108.15', ..., '346.45', '306.6', '6844.5'],
      dtype=object)
```

pd.to_numeric(df.TotalCharges)

pd.to_numeric(df.TotalCharges,errors='coerce').isnull()

```
0       False
1       False
2       False
3       False
4       False
        ...
7038    False
7039    False
7040    False
7041    False
7042    False
Name: TotalCharges, Length: 7043, dtype: bool
```

df[pd.to_numeric(df.TotalCharges,errors='coerce').isnull()]

This error tells that some values are not string they are empty --> " " at position 488

df[df.TotalCharges==' '].shape

(11, 20)

df1 = df[df.TotalCharges!=' ']

df1.TotalCharges = pd.to_numeric(df1.TotalCharges)

df1.TotalCharges.values

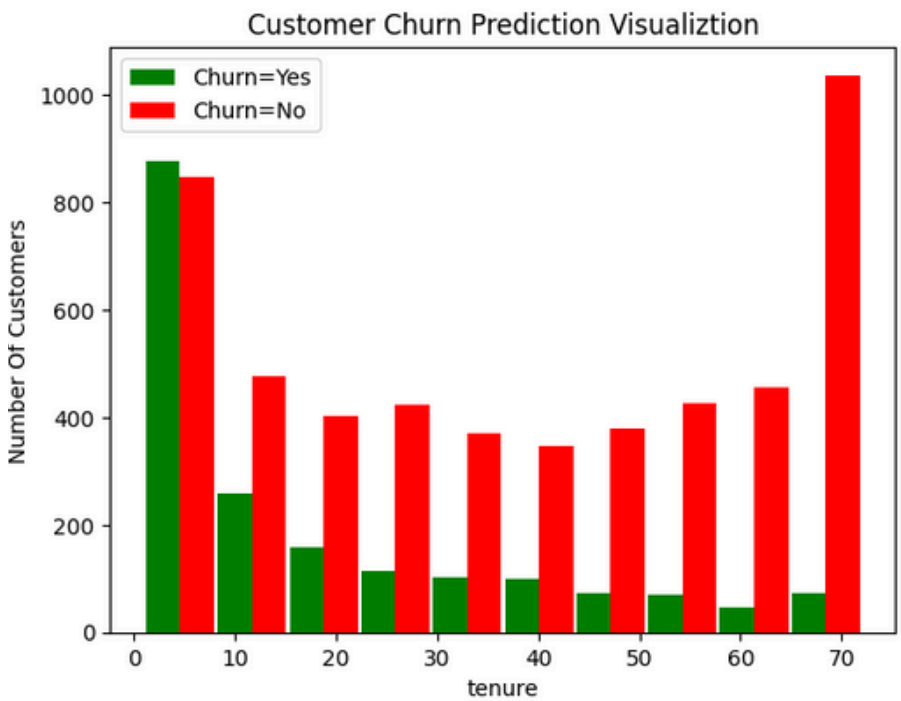
```
array([  29.85, 1889.5 ,  108.15, ...,  346.45,  306.6 , 6844.5 ])
```

tenure_churn_no = df1[df1.Churn=='No'].tenure
tenure_churn_yes = df1[df1.Churn=='Yes'].tenure

plt.xlabel("tenure")
plt.ylabel("Number Of Customers")
plt.title("Customer Churn Prediction Visualiztion")

blood_sugar_men = [113, 85, 90, 150, 149, 88, 93, 115, 135, 80, 77, 82, 129]
blood_sugar_women = [67, 98, 89, 120, 133, 150, 84, 69, 89, 79, 120, 112, 100]

plt.hist([tenure_churn_yes, tenure_churn_no], rwidth=0.95, color=['green','red'],label=['Churn=Yes','Churn=No'])
plt.legend()



```
def uniue_values(df1):
   for col in df1:
      if df1[col].dtype=='object':
         print(f"{col} : {df1[col].unique()}")
uniue_values(df1)
```

```
gender : ['Female' 'Male']
Partner : ['Yes' 'No']
Dependents : ['No' 'Yes']
PhoneService : ['No' 'Yes']
MultipleLines : ['No phone service' 'No' 'Yes']
InternetService : ['DSL' 'Fiber optic' 'No']
OnlineSecurity : ['No' 'Yes' 'No internet service']
OnlineBackup : ['Yes' 'No' 'No internet service']
DeviceProtection : ['No' 'Yes' 'No internet service']
TechSupport : ['No' 'Yes' 'No internet service']
StreamingTV : ['No' 'Yes' 'No internet service']
StreamingMovies : ['No' 'Yes' 'No internet service']
Contract : ['Month-to-month' 'One year' 'Two year']
PaperlessBilling : ['Yes' 'No']
PaymentMethod : ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']
Churn : ['No' 'Yes']
```

```
df1.replace('No internet service','No',inplace=True)
 df1.replace('No phone service','No',inplace=True)
```

```
yes_no_columns =
['Partner','Dependents','PhoneService','MultipleLines','OnlineSecurity','OnlineBackup',
            'DeviceProtection','TechSupport','StreamingTV','StreamingMovies','PaperlessBilling','Churn']
for col in yes_no_columns:
   df1[col].replace({'Yes': 1,'No': 0},inplace=True)
```

```
for col in df1:
   print(f'{col}: {df1[col].unique()}')
```

```
gender: ['Female' 'Male']
SeniorCitizen: [0 1]
Partner: [1 0]
Dependents: [0 1]
tenure: [ 1 34  2 45  8 22 10 28 62 13 16 58 49 25 69 52 71 21 12 30 47 72 17 27
  5 46 11 70 63 43 15 60 18 66  9  3 31 50 64 56  7 42 35 48 29 65 38 68
 32 55 37 36 41  6  4 33 67 23 57 61 14 20 53 40 59 24 44 19 54 51 26 39]
PhoneService: [0 1]
MultipleLines: [0 1]
InternetService: ['DSL' 'Fiber optic' 'No']
OnlineSecurity: [0 1]
OnlineBackup: [1 0]
DeviceProtection: [0 1]
TechSupport: [0 1]
StreamingTV: [0 1]
StreamingMovies: [0 1]
Contract: ['Month-to-month' 'One year' 'Two year']
PaperlessBilling: [1 0]
PaymentMethod: ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']
MonthlyCharges: [29.85 56.95 53.85 ... 63.1  44.2  78.7 ]
TotalCharges: [  29.85 1889.5   108.15 ...  346.45  306.6  6844.5 ]
Churn: [0 1]
```

```
df1['gender'].replace({'Female':1,'Male':0},inplace=True)
```

6. Convert columns to the correct data types as needed.

```
df2 = pd.get_dummies(data=df1, columns=['InternetService','Contract','PaymentMethod'])
df2 = df2.astype(int)
df2.columns
```

```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'OnlineSecurity', 'OnlineBackup',
       'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies',
       'PaperlessBilling', 'MonthlyCharges', 'TotalCharges', 'Churn',
       'InternetService_DSL', 'InternetService_Fiber optic',
       'InternetService_No', 'Contract_Month-to-month', 'Contract_One year',
       'Contract_Two year', 'PaymentMethod_Bank transfer (automatic)',
       'PaymentMethod_Credit card (automatic)',
       'PaymentMethod_Electronic check', 'PaymentMethod_Mailed check'],
      dtype='object')
```

```
cols_to_scale = ['tenure','MonthlyCharges','TotalCharges']

from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df2[cols_to_scale] = scaler.fit_transform(df2[cols_to_scale]) # converts the values in the
range of 0 to 1
```

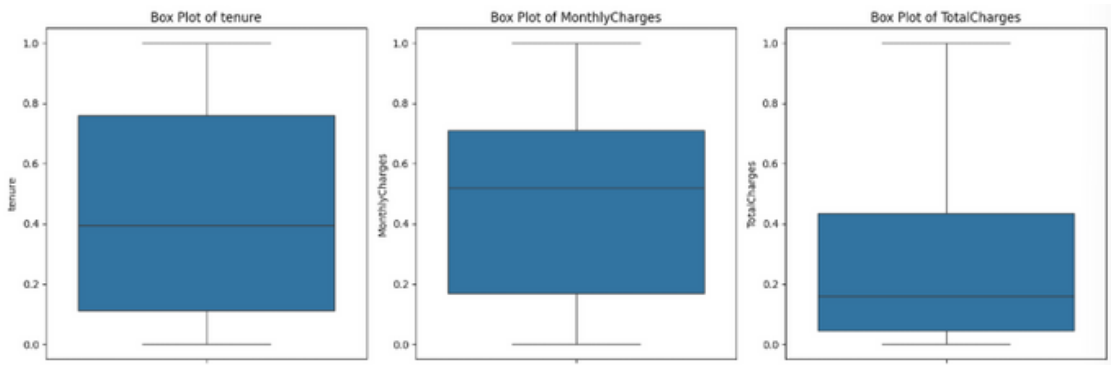
	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	OnlineSecurity	OnlineBackup	DeviceProte
0	1	0	1	0	0.000000	0	0	0	1	
1	0	0	0	0	0.464789	1	0	1	0	
2	0	0	0	0	0.014085	1	0	1	1	
3	0	0	0	0	0.619718	0	0	1	0	
4	1	0	0	0	0.014085	1	0	0	0	

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Convert TotalCharges to numeric
df2['TotalCharges'] = pd.to_numeric(df2['TotalCharges'], errors='coerce')

# Create box plots for numerical features
num_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']

# Plotting box plots
plt.figure(figsize=(15, 5))
for i, col in enumerate(num_cols):
    plt.subplot(1, 3, i+1)
    sns.boxplot(y=col, data=df2)
    plt.title(f'Box Plot of {col}')
    plt.tight_layout()
plt.show()
```



8. Perform feature engineering, creating new features that may be relevant to predicting customer churn.

```
df3=df2

# Create a new feature: 'tenure_group'
df3['tenure_group'] = pd.cut(df3['tenure'], bins=[0, 12, 24, 48, 60, 72],
                             labels=['0-12', '13-24', '25-48', '49-60', '61-72'])

# Create interaction features if useful, e.g., Monthly*Tenure
df3['LifetimeCharges'] = df3['MonthlyCharges'] * df3['tenure']
```

9. Normalize or scale the data if necessary.

Data is Standardise and normalised for building model above

10. Split the dataset into training and testing sets for further analysis.

```
x = df2.drop('Churn',axis='columns')
y = df2['Churn']

from sklearn.model_selection import train_test_split

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=5)
```

11. Export the cleaned dataset for future analysis or modeling.

```
df_cleaned = x.copy()
df_cleaned['Churn'] = y
df_cleaned.to_csv("Cleaned_Telecom_Customer_Churn.csv", index=False)
```